CS 486/686
Training Models From Scratch

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Lecture 17

Core loop: data → model → loss → gradient → update

Recorded lecture (asynchronous)

🎥 This lecture is **pre-recorded** - I'm away at **ICML** this week, so there's no in-person class today. Watch at your own pace.

📅 **Due today (Tue Jul 7):** Chat 8. The CS686 project proposal is now due **Thu Jul 9** (submit on LEARN under "Project Proposal").

💬 Questions? Post on **Piazza** - the TAs are available, and I'll follow up when I'm back.

Today's target

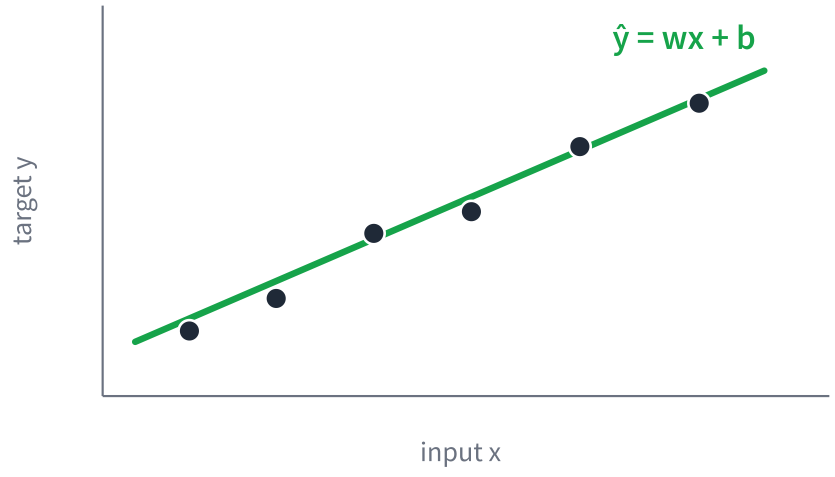
By the end, this loop should feel readable:

```
for x, y in loader:
    pred = model(x)
    loss = loss_fn(pred, y)
    loss.backward()
    optimizer.step()
    optimizer.zero_grad()
```

Everything later - neural nets, LLMs, diffusion - scales up this idea.

The learning recipe

Parameters are the knobs

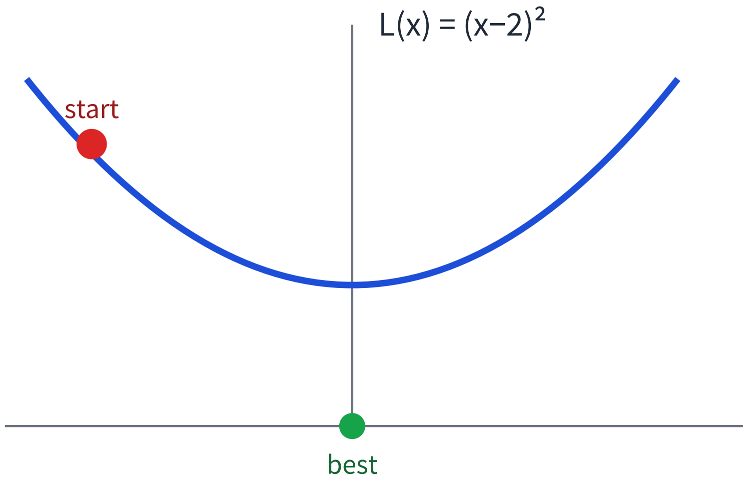


The data is fixed.

The model has tunable **parameters**: w and b.

Training means choosing parameters that make predictions good.

Start with one knob



Forget ML for one slide.

We only need to make a loss small.

This is the smallest possible version of training.

Move downhill

The gradient points uphill, so step the other way.

$$x \leftarrow x - \eta \frac{dL}{dx}$$

$-dL/dx$

direction that decreases loss

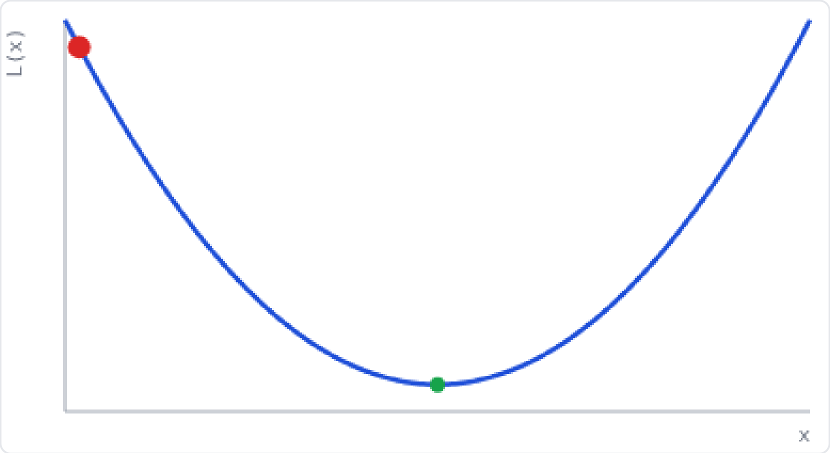
η

learning rate: how big a step?

Playground: gradient descent LIVE

Interactive demo — runs live in your browser (open the slides to try it).

learning rate η 0.10 Step Run Reset




```
step: 0
x = -3.000
L(x) = 25.000
dL/dx = -10.000
```

Small η: slow but steady. Around η = 1 it overshoots; push higher and it diverges.

PyTorch computes gradients for us

Manual derivative

```
def loss(x):
    return (x - 2) ** 2

def manual_grad(x):
    return 2 * (x - 2)
```

Automatic differentiation

```
x = nn.Parameter(torch.tensor(-4.0))
L = (x - 2) ** 2
L.backward()
print(x.grad)  # -12
```

For millions of parameters, we rely on autograd. But how does it know the answer?

How can gradients be automatic?

Every operation records its own **local derivative**; the **chain rule** multiplies them backward.

Forward: $u = x - 2$, then $L = u^2$.

Chain rule: $\frac{dL}{dx} = \frac{dL}{du} \cdot \frac{du}{dx}$

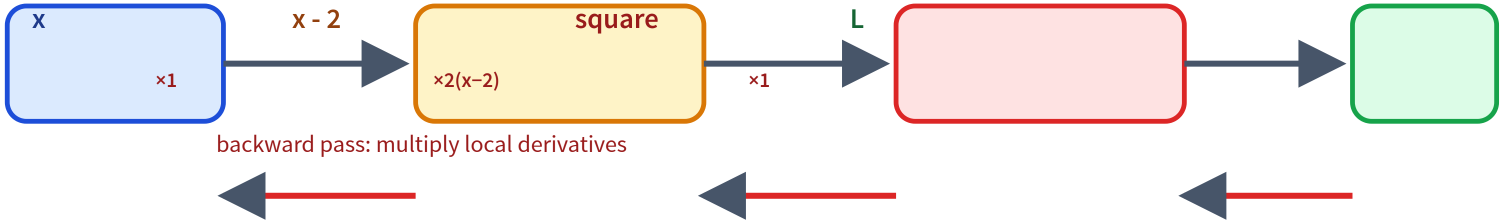
Local: $\frac{dL}{du} = 2u$ (since $L = u^2$)

Local: $\frac{du}{dx} = 1$ (since $u = x - 2$)

Multiply: $\frac{dL}{dx} = 2u \cdot 1 = 2(x - 2)$

At $x = -4$: $\frac{dL}{dx} = 2(-6) = -12$ — exactly `x.grad`.

Autograd follows the computation graph



Run it: autograd from scratch LIVE PYTHON

Interactive demo — runs live in your browser (open the slides to try it).

Run Python

Reset code

```
# A tiny reverse-mode autograd (this is what .b
class Value:
    def __init__(self, data, _children=()):
        self.data = data
        self.grad = 0.0
        self._backward = lambda: None
        self._prev = set(_children)

    def __sub__(self, other):
        other = other if isinstance(other, Valu
        out = Value(self.data - other.data, (se
        def _backward():
```

```
Click "Run Python" to execute.
```

Real Python in your browser. The graph x → (x−2) → square → L is built as you compute, then backward() multiplies local derivatives.

From one knob to many

A model has many parameters, collected as θ. The loop does not change.

$$\theta \leftarrow \theta - \eta \, \nabla_\theta L(\theta)$$

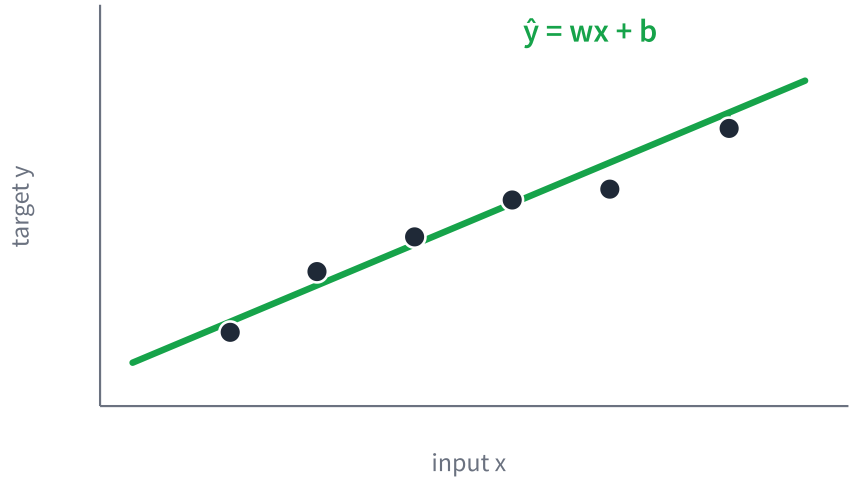
```
optimizer = SGD(model.parameters(), lr=0.1)

for x, y in loader:
    pred = model(x)
    loss = loss_fn(pred, y)
    loss.backward()
    optimizer.step()
    optimizer.zero_grad()
```

`model.parameters()` → the parameters θ

`loss.backward()` → the gradient $\nabla_\theta L$

`optimizer.step()` → update $\theta \leftarrow \theta - \eta \nabla_\theta L$

Example 1: regression



Prediction: $\hat{y} = wx + b$ (w, b are the parameters)

Error: $L = \frac{1}{m}\sum_i(\hat{y}_i - y_i)^2$

Start with a bad line — big errors (dashed).

Measure the error, nudge w, b to shrink it.

Repeat — the errors keep shrinking.

Good fit: the errors are small.

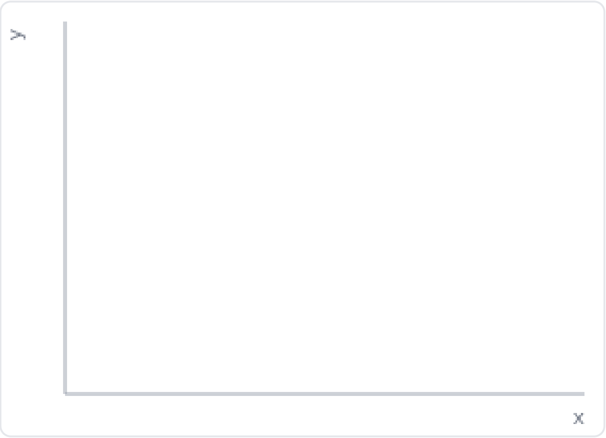
Regression in PyTorch

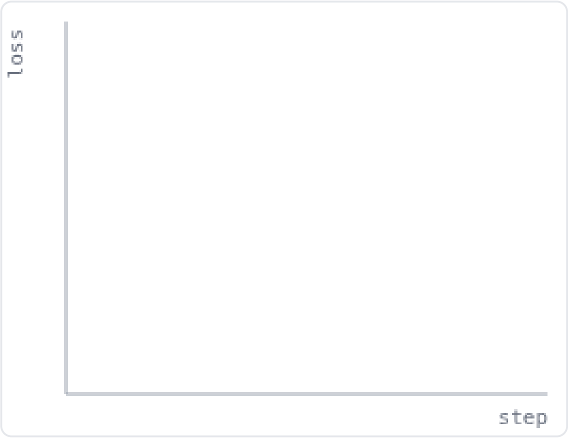
```python
# the knob: a line y = wx + b
model = nn.Linear(1, 1)
# how we score error: mean squared error
loss_fn = nn.MSELoss()
# how we update the parameters
optimizer = torch.optim.SGD(model.parameters(), lr=0.1)
# forward: predict from inputs
pred = model(x)
# measure how wrong we are
loss = loss_fn(pred, y)
# autograd fills in the gradients
loss.backward()
# one step downhill: update w and b
optimizer.step()
# clear gradients before the next batch
optimizer.zero_grad()
```

Run it: train the line LIVE PYTHON

Interactive demo — runs live in your browser (open the slides to try it).

learning rate 0.10 steps 60 Train






```
Click "Train" to run the loop in Python.
```

Same loop as the slide, in numpy: predict → MSE → gradients → update. Watch the line snap to the data as the loss drops. High learning rate can overshoot.

Example 2: classification

Now the target is a **category**, not a number.

Regression

$\hat{y} = 217.3$

a number

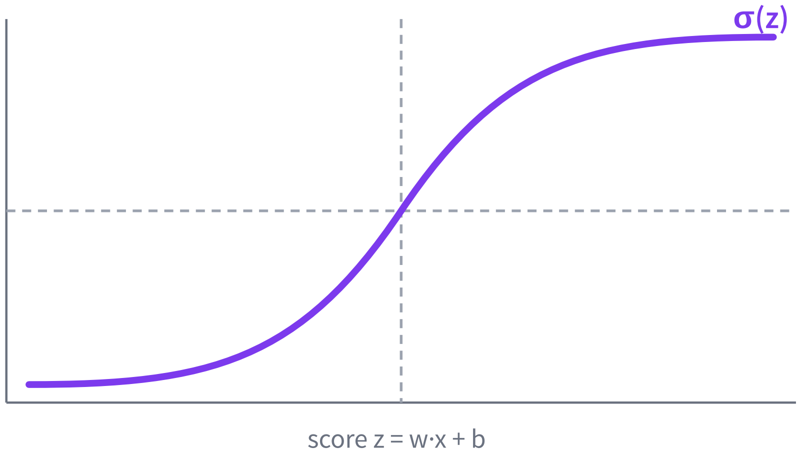
Classification

$\hat{y} = \text{“spam”}$

a class

But gradient descent needs a smooth output — so we predict a **probability** instead.

Logistic regression: predict a probability



Compute a score:

$$z = w \cdot x + b$$

Squash it into a probability:

$$P(\text{spam} \mid x) = \sigma(z) = \frac{1}{1+e^{-z}}$$

Loss for classification

Cross-entropy punishes being confidently wrong.

Recall $\hat{y} = P(y = 1 \mid x) = \sigma(z)$ — the probability the model predicts.

$$L = -\Big[y \log \hat{y} + (1 - y)\log(1 - \hat{y})\Big]$$

True label spam $\Rightarrow y = 1$, so $L = -\log \hat{y}$.

confident & right

$\hat{y} = 0.9$

$L = -\log 0.9 \approx 0.11$

confident & wrong

$\hat{y} = 0.1$

$L = -\log 0.1 \approx 2.30$

Many classes: softmax

One score per class; softmax turns scores into probabilities.

$$P(y = k \mid x) = \frac{e^{z_k}}{\sum_j e^{z_j}}$$

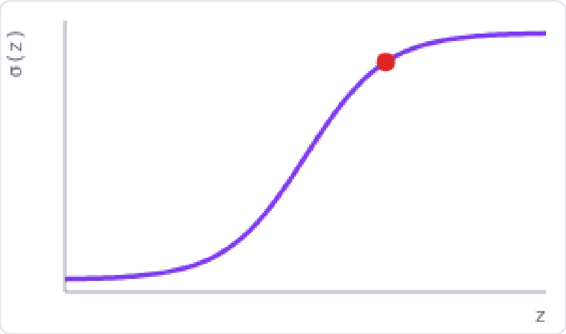
Next-token prediction is exactly this, over a vocabulary.

Playground: losses & softmax LIVE

Interactive demo — runs live in your browser (open the slides to try it).

score z 2.0

flip true label




```
true label: spam (y=1)
ŷ = σ(z) = 0.881
loss = 0.127
confident & right
```

Cross-entropy = −log(prob of the true class). Confident and wrong is punished hard.

temperature 1.0

dog 1.0

cat 3.0

bird 1.5

fish 0.5

Softmax turns scores into probabilities. Low temperature sharpens; high temperature flattens (this is the decoding knob in L20/L21).

The full loop, annotated

```
# loop over the training data
for x, y in loader:
# forward pass: predict
    pred = model(x)
# measure the error
    loss = loss_fn(pred, y)
# backward pass: compute gradients
    loss.backward()
# update the parameters
    optimizer.step()
# reset gradients for the next step
    optimizer.zero_grad()
```

This is the pattern to recognize for the rest of the module.

Why mini-batches?

Full batch — entire dataset → **1** update / epoch

Mini-batch — batch | batch | batch | batch → **4** updates / epoch

Full batch: few, expensive, smooth steps. Mini-batch: many, cheap, noisy steps — usually faster progress, and GPU-friendly.

Split the data: train / validation / test

train

updates the weights

validation

chooses settings

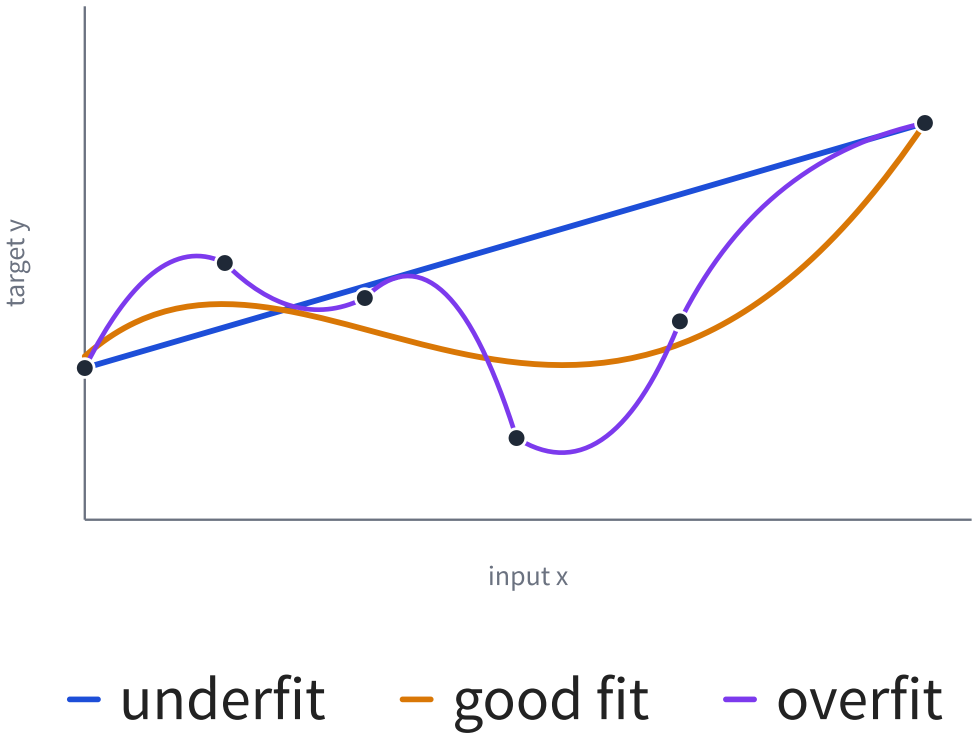
test

final, one-time estimate

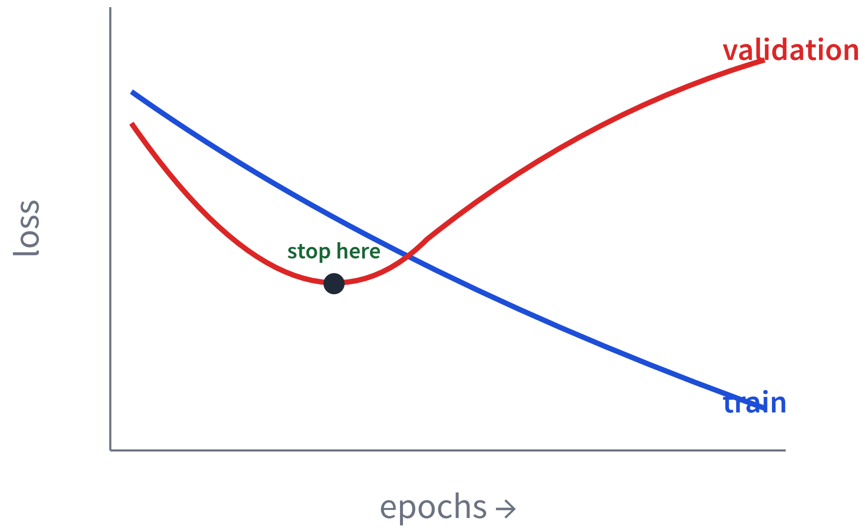
Never tune on the test set.

More power is not always better

Same data, increasingly flexible models.



Watch train and validation loss



Training loss keeps dropping.

Validation loss starts **rising** — that growing gap is overfitting. Stop where validation bottoms out.

Playground: overfitting LIVE PYTHON

Interactive demo — runs live in your browser (open the slides to try it).

polynomial degree 1 ridge λ 0.00 Fit in Python

Blue = training points, red = validation points. Raise the degree: training loss keeps dropping but validation loss turns up - overfitting. Ridge λ pulls it back.

Three ways to fight overfitting

More data

show the model
more of the world

Regularize

prefer simpler
weights

$$L(\theta) + \lambda||\theta||^2$$

Early stop

stop when validation
turns up

What carries forward?

Only the **model** and the **loss** change. The loop stays the same.

Task	Model (architecture)	Loss
Regression (today)	Linear / MLP	squared error
Language model	Transformer	cross-entropy (next token)
Image generation	Diffusion U-Net	squared error (predict noise)

forward → loss → backward → step, every time.

Recap - next: learned representations

✓ A model learns by minimizing a **loss**.

✓ **Autograd** computes gradients through a graph.

✓ Regression uses squared error; classification uses **cross-entropy**.

✓ Validation loss catches **overfitting**.

L18: if linear models use fixed features, how do neural nets learn their own?